

The Exponent Rules (Part IV)

Problems Requiring More than Four Rules

Simplify each expression below and write the Roman numerals of the exponent rules that you used in the margin. Do not leave your answers with fraction exponents or negative exponents.

① $(2x^4y^2 \cdot y^3)^{\frac{1}{2}} (-5x^4y^{\frac{1}{2}} \cdot x^{-6})^3$

I $(2x^4y^{2+3})^{\frac{1}{2}} (-5x^{4+(-6)}y^{\frac{1}{2}})^3$

$(2x^4y^5)^{\frac{1}{2}} (-5x^{-2}y^{\frac{1}{2}})^3$

III $2^{\frac{1}{2}}(x^4)^{\frac{1}{2}}(y^5)^{\frac{1}{2}}(-5)^3(x^{-2})^3(y^{\frac{1}{2}})^3$

IV, V $\sqrt{2} x^{4(\frac{1}{2})} y^{5(\frac{1}{2})} (-125) x^{-2(3)} y^{\frac{1}{2}(3)}$

$-125x^2y^{5/2}x^{-6}y^{3/2}\sqrt{2}$

I $-125x^{2+(-6)}y^{5/2+3/2}\sqrt{2}$

$-125x^{-4}y^4\sqrt{2}$

VIII

$$\boxed{\frac{-125y^4\sqrt{2}}{x^4}}$$

② $(-3a^6c^{\frac{2}{5}} \cdot a)^2 (-2b^{\frac{3}{7}} \cdot b^{\frac{4}{7}}a^8)^4 (2b)^2$

VI, I, III $(-3a^{6+1}c^{2/5}a^1)^2 (-2b^{\frac{3}{7}+\frac{4}{7}}a^8)^4 (2^2b^2)$

I $(-3a^{6+1}c^{2/5}a^1)^2 (-2b^1a^8)^4 (4b^2)$

VI $(-3a^7c^{2/5}a^1)^2 (-2b^1a^8)^4 (4b^2)$

III $(-3)^2(a^7)^2(c^{2/5})^2(-2)^4b^4(a^8)^4(4b^2)$

V $9a^{7(2)}c^{2/5(2)}(16)b^4a^{8(4)}(4b^2)$

$9a^{14}c^{4/5}(16)b^4a^{32}(4b^2)$

I, X $576a^{14+32}(\sqrt[5]{c})^4b^{4+2}$

$$\boxed{576a^{46}b^6(\sqrt[5]{c})^4}$$

$$\textcircled{3} (4a^5 \cdot a^6 \cdot b)^2 (7^{-1} b^{\frac{2}{3}} \cdot a \cdot b^{\frac{1}{3}})^2$$

$$\text{I} \quad (4a^{5+6}b)^2 (7^{-1}b^{\frac{2}{3}+\frac{1}{3}}a)^2$$

$$(4a^{11}b)^2 (7^{-1}b^1a)^2$$

$$\text{III} \quad 4^2(a^{11})^2b^2(7^{-1})^2(b^1)^2a^2$$

$$\text{IV} \quad 16a^{22}b^27^{-2}b^2a^2$$

$$16a^{22}b^27^{-2}b^2a^2$$

$$\text{I} \quad 16a^{22+2}b^{2+2}(7^{-2})$$

$$16a^{24}b^4(7^{-2})$$

$$\text{VIII} \quad \frac{16a^{24}b^4}{7^2}$$

$$\boxed{\frac{16a^{24}b^4}{49}}$$

$$\textcircled{4} (-4ab)(a^3b \cdot b^7)^6(3a^2 \cdot a^{-5}c^{\frac{1}{4}})^3$$

$$\text{VI, I} \quad (-4ab)(a^3b^1 \cdot b^7)^6(3a^{2+(-5)}c^{\frac{1}{4}})^3$$

$$\text{I} \quad (-4ab)(a^3b^{1+7})^6(3a^{-3}c^{\frac{1}{4}})^3$$

$$\text{III} \quad (-4ab)(a^3b^8)^6(3^3)(a^{-3})^3(c^{\frac{1}{4}})^3$$

$$\text{III, V} \quad (-4ab)(a^3)^6(b^8)^6(27)a^{-3(3)}c^{\frac{1}{4}(3)}$$

$$\text{V} \quad (-4ab)a^{3(6)}b^{8(6)}(27)a^{-9}c^{\frac{3}{4}}$$

$$\text{VI} \quad (-4a^1b^1)a^{18}b^{48}(27)a^{-9}c^{\frac{3}{4}}$$

$$\text{I} \quad -108a^{1+18+(-9)}b^{1+48}c^{\frac{3}{4}}$$

$$\text{X} \quad \boxed{-108a^{10}b^{49}(\sqrt[4]{c})^3}$$

$$⑤ \left[(5x^3)^2 \left(\frac{3y^7}{x^{-4}a^{\frac{1}{7}}} \right)^{-3} \right]^2$$

III, IV

$$\left[5^2 (x^3)^2 \cdot \frac{(3y^7)^{-3}}{(x^{-4}a^{\frac{1}{7}})^{-3}} \right]^2$$

V, III

$$\left[25 x^{3(2)} \cdot \frac{3^{-3} (y^7)^{-3}}{(x^{-4})^{-3} (a^{\frac{1}{7}})^{-3}} \right]^2$$

VIII, V

$$\left[25 x^6 \cdot \frac{y^{7(-3)}}{3^3 x^{-4(-3)} a^{\frac{1}{7}(-3)}} \right]^2$$

$$\left[25 x^6 \cdot \frac{y^{-21}}{27 x^{12} a^{-\frac{3}{7}}} \right]^2$$

VIII, VIII

$$\left[\frac{25 x^6}{1} \cdot \frac{a^{\frac{3}{7}}}{27 x^{12} y^{21}} \right]^2$$

$$\left[\frac{25 x^6 a^{\frac{3}{7}}}{27 x^{12} y^{21}} \right]^2$$

II

$$\left[\frac{25 x^{6-12} a^{\frac{3}{7}}}{27 y^{21}} \right]^2$$

$$\left[\frac{25 x^{-6} a^{\frac{3}{7}}}{27 y^{21}} \right]^2$$

VIII

$$\left[\frac{25 a^{\frac{3}{7}}}{27 y^{21} x^6} \right]^2$$

IV

$$\frac{(25 a^{\frac{3}{7}})^2}{(27 y^{21} x^6)^2}$$

III

$$\frac{(25^2)(a^{\frac{3}{7}})^2}{(27^2)(y^{21})^2(x^6)^2}$$

V

$$\frac{625 a^{\frac{3}{7}(2)}}{729 y^{21(2)} x^{6(2)}}$$

$$\frac{625 a^{\frac{6}{7}}}{729 y^{42} x^{12}}$$

V

$$\boxed{\frac{625(\sqrt[7]{a})^6}{729 y^{42} x^{12}}}$$

$$\textcircled{6} \left[\left(\frac{b^{-1} a^{\frac{2}{3}}}{2c} \right)^6 (-3ab)^2 \right]^3$$

IV, III

$$\left[\frac{(b^{-1} a^{\frac{2}{3}})^6}{(2c)^6} (-3)^2 a^2 b^2 \right]^3$$

III

$$\left[\frac{(b^{-1})^6 (a^{\frac{2}{3}})^6}{2^6 c^6} \cdot 9 a^2 b^2 \right]^3$$

V

$$\left[\frac{b^{-1(6)} a^{\frac{2}{3}(6)}}{64 c^6} \cdot 9 a^2 b^2 \right]^3$$

$$\left[\frac{b^{-6} a^4}{64 c^6} \cdot \frac{9 a^2 b^2}{1} \right]^3$$

I

$$\left[\frac{9 b^{-6+2} a^{4+2}}{64 c^6} \right]^3$$

VIII

$$\left[\frac{9b^{-4}a^6}{64c^6} \right]^3$$

IV

$$\left[\frac{9a^6}{64c^6b^4} \right]^3$$

$$\frac{(9a^6)^3}{(64c^6b^4)^3}$$

III

$$\frac{9^3(a^6)^3}{64^3(c^6)^3(b^4)^3}$$

V

$$\frac{729a^{6(3)}}{262,144c^{6(3)}b^{4(3)}}$$

$$\boxed{\frac{729a^{18}}{262,144c^{18}b^{12}}}$$

$$\textcircled{7} \left(\frac{3x^{-3}y^4}{2a^5b^0} \right)^4 \left(\frac{x^{-2}a^3}{2x^2} \right)^5$$

VII, VIII
II

$$\left(\frac{3y^4}{2x^3a^5(1)} \right)^4 \left(\frac{x^{-2-2}a^3}{2} \right)^5$$

IV

$$\frac{(3y^4)^4}{(2x^3a^5)^4} \left(\frac{x^{-4}a^3}{2} \right)^5$$

III, VIII

$$\frac{3^4(y^4)^4}{2^4(x^3)^4(a^5)^4} \left(\frac{a^3}{2x^4} \right)^5$$

VI, IV

$$\frac{81y^{4(4)}}{16x^{3(4)}a^{5(4)}} \cdot \frac{(a^3)^5}{(2x^4)^5}$$

VI, III

$$\frac{81y^{16}}{16x^{12}a^{20}} \cdot \frac{a^{3(5)}}{2^5(x^4)^5}$$

V

$$\frac{81y^{16}}{16x^{12}a^{20}} \cdot \frac{a^{15}}{32x^{4(5)}}$$

$$\frac{81y^{16}}{16x^{12}a^{20}} \cdot \frac{a^{15}}{32x^{20}}$$

I

$$\frac{81y^{16}a^{15}}{512x^{12+20}a^{20}}$$

II

$$\frac{81y^{16}a^{15-20}}{512x^{32}}$$

$$\frac{81y^{16}a^{-5}}{512x^{32}}$$

VIII

$$\boxed{\frac{81y^{16}}{512x^{32}a^5}}$$

$$\textcircled{8} \left(\frac{4x^{-4}y^6}{y^5c^{-3}} \right)^2 \left(\frac{y^3a^5x^{-2}}{4a^{-10}} \right)^3$$

II, VIII

VIII

$$\left(\frac{4y^{6-5}c^3}{x^4} \right)^2 \left(\frac{y^3a^{5-(-10)}}{4x^2} \right)^3$$

$$\left(\frac{4y^1c^3}{x^4} \right)^2 \left(\frac{y^3a^{15}}{4x^2} \right)^3$$

VI, IV

$$\frac{(4yc^3)^2}{(x^4)^2} \cdot \frac{(y^3a^{15})^3}{(4x^2)^3}$$

III, V

$$\frac{4^2y^2(c^3)^2}{x^{4(2)}} \cdot \frac{(y^3)^3(a^{15})^3}{4^3(x^2)^3}$$

V

$$\frac{16y^2c^{3(2)}}{x^8} \cdot \frac{y^{3(3)}a^{15(3)}}{64x^{2(3)}}$$

$$\frac{16y^2c^6}{x^8} \cdot \frac{y^9a^{45}}{64x^6}$$

I

$$\frac{16y^{2+9}c^6a^{45}}{64x^{8+6}}$$

$$\frac{16y^{11}c^6a^{45}}{64x^{14}}$$

$$\boxed{\frac{y^{11}c^6a^{45}}{4x^{14}}}$$

$$\textcircled{9} \frac{(a^3b^4)^{-1}}{(3b^2a^7)^3} \cdot \left(\frac{5a^2b^4}{4a^{-8}b^2}\right)^3 \div \left(\frac{a^{-2}}{a^0b^{-5}}\right)^4$$

III, II

VIII, VIII'

VII

$$\frac{(a^3)^{-1}(b^4)^{-1}}{3^3(b^2)^3(a^7)^3} \cdot \left(\frac{5a^{2-(-8)}b^{4-2}}{4}\right)^3 \div \left(\frac{b^5}{1 \cdot a^2}\right)^4$$

V, IV

$$\frac{a^{3(-1)}b^{4(-1)}}{27b^{2(3)}a^{7(3)}} \cdot \left(\frac{5a^{10}b^2}{4}\right)^3 \div \frac{(b^5)^4}{(a^2)^4}$$

IV, V

$$\frac{a^{-3}b^{-4}}{27b^6a^{21}} \cdot \frac{(5a^{10}b^2)^3}{4^3} \div \frac{b^{5(4)}}{a^{2(4)}}$$

II, III

$$\frac{a^{-3-21}b^{-4-6}}{27} \cdot \frac{5^3(a^{10})^3(b^2)^3}{64} \div \frac{b^{20}}{a^8}$$

V

$$\frac{a^{-24}b^{-10}}{27} \cdot \frac{125a^{10(3)}b^{2(3)}}{64} \cdot \frac{a^8}{b^{20}}$$

VIII

$$\frac{1}{27a^{24}b^{10}} \cdot \frac{125a^{30}b^6}{64} \cdot \frac{a^8}{b^{20}}$$

I

$$\frac{125a^{30+8}b^6}{1728a^{24}b^{10+20}}$$

$$\frac{125a^{38}b^6}{1728a^{24}b^{30}}$$

II

$$\frac{125a^{38-24}b^{6-30}}{1728}$$

$$\frac{125a^{14}b^{-24}}{1728}$$

VIII

$$\boxed{\frac{125a^{14}}{1728b^{24}}}$$

PEMDAS

$$\textcircled{10} \left(\frac{x^3c^{-4}}{c^2y^0x^8b^{-3}} \right)^3 \div \frac{(y^2c^4x)^2}{(c^3x^0y^5)^3} \cdot \frac{b^2}{x^4}$$

II, VII

VIII, III

$$\left(\frac{x^{3-8}c^{-4-2}b^3}{1} \right)^3 \div \frac{(y^2)^2(c^4)^2x^2}{(c^3)^3(1)^3(y^5)^3} \cdot \frac{b^2}{x^4}$$

V

$$(x^{-5}c^{-6}b^3)^3 \div \frac{y^{2(2)}c^{4(2)}x^2}{c^{3(3)}(1)y^{5(3)}} \cdot \frac{b^2}{x^4}$$

III

$$(x^{-5})^3(c^{-6})^3(b^3)^3 \div \frac{y^4c^8x^2}{c^9y^{15}} \cdot \frac{b^2}{x^4}$$

V, II

$$x^{-5(3)}c^{-6(3)}b^{3(3)} \div y^{4-15}c^{8-9}x^2 \cdot \frac{b^2}{x^4}$$

$$x^{-15}c^{-18}b^9 \div y^{-11}c^{-1}x^2 \cdot \frac{b^2}{x^4}$$

$$\frac{x^{-15} c^{-18} b^9}{1} \div \frac{y^{-11} c^{-1} x^2}{1} \cdot \frac{b^2}{x^4}$$

$$\frac{x^{-15} c^{-18} b^9}{1} \cdot \frac{1}{y^{-11} c^{-1} x^2} \cdot \frac{b^2}{x^4}$$

$$\frac{x^{-15} c^{-18} b^{9+2}}{y^{-11} c^{-1} x^{2+4}}$$

$$\frac{x^{-15} c^{-18} b^{11}}{y^{-11} c^{-1} x^6}$$

$$\frac{x^{-15-6} c^{-18-(-1)} b^{11}}{y^{-11}}$$

$$\frac{x^{-21} c^{-17} b^{11}}{y^{-11}}$$

VIII, VIII

$$\frac{b^{11} y^{11}}{x^{21} c^{17}}$$

II